

Message Authentication

◆ Introduction:

Message authentication is a security process used to ensure that a message is **genuine, unaltered, and sent by an authorized sender**. It protects communication over insecure networks from attacks like modification, impersonation, and replay.

◆ Need for Message Authentication:

Message authentication is essential due to the following reasons:

1. **Data Integrity**
 - Ensures that the message is not modified during transmission.
 2. **Authentication of Sender**
 - Confirms that the message is sent by a legitimate sender.
 3. **Protection Against Unauthorized Modification**
 - Prevents attackers from altering the message.
 4. **Protection Against Replay Attacks**
 - Ensures old messages cannot be reused by attackers.
 5. **Non-repudiation (in some techniques)**
 - Sender cannot deny sending the message.
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◆ Techniques for Message Authentication:

The commonly used techniques are:

1. **Message Authentication Code (MAC)**
 - Uses a secret key to generate a tag for authentication.
 2. **Hash Functions**
 - Converts message into a fixed-size hash value.
 3. **Digital Signatures**
 - Uses public key cryptography for authentication and integrity.
 4. **Encryption-based Authentication**
 - Message is encrypted to ensure authenticity.
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◆ Digital Signature (Explanation):

👉 Definition:

A **Digital Signature** is a cryptographic technique used to verify the **authenticity, integrity, and non-repudiation** of a message using a pair of keys (private and public keys).

👉 Working:

1. Sender prepares the message
 2. A **hash function** is applied to generate a message digest
 3. The digest is encrypted using the sender's **private key**
 4. This encrypted digest is called the **digital signature**
 5. The message and signature are sent to the receiver
 6. Receiver decrypts the signature using sender's **public key**
 7. Receiver generates a new hash of the received message
 8. Both hashes are compared:
 - If same → message is authentic
 - If different → message is altered
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👉 Advantages:

- Ensures **data integrity**
 - Provides **authentication of sender**
 - Offers **non-repudiation**
 - Widely used in secure communications (email, e-commerce)
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